

Same SAM2-propagation backbone, re-worked for *C. elegans* neurites

Grey = shared spine WORM = added or changed here NEW added CHANGED reworked

LIVER pipeline		WORM pipeline	
Xing et al. (<i>sam2maskpropagator</i>)		this work	
Input	SWC morphological reconstruction (μm)	CHANGED	CATMAID skeleton, aligned to the raw stack by a per-section registration
Structure	one compact object per prompt	NEW	branching neuron tree sliced into maximal linear chains (DFS + inserted virtual nodes)
Anchor prompt	single positive point on the mid-slice	CHANGED	positive node + negative points on neighbouring neurons; node-anchored multimask selection
SAM2 image mode → largest connected component → bounding box			
Resolution	fixed crop (2000 px; 6000 px hepatocytes)	CHANGED	downscale-8 to locate, then re-predict on a tiered high-resolution crop
SAM2 video mode: bidirectional propagation from the anchor			
Quality control	not in the released SAM2 code	NEW	inline QC + flagging: area ratio, temporal IoU, skeleton containment, predicted-IoU
Orchestration	linear loop over objects	CHANGED	batch runner with manifest, crash-resume, and a triage queue
Correction	manual proofreading (unelaborated)	NEW	napari human-in-the-loop review: correct only flagged frames, re-predict / resume over live state
Composite	instance volume, last-writer-wins	CHANGED	first-writer-wins; optional non-overlap co-propagation experiment
Evaluation	morphometry (PyRadiomics); no segmentation-accuracy metric	NEW	cross-worm ground truth: per-neuron ERL + VOI _{split/merge} + ARAND
per-object instance masks → Blender			

The liver pipeline also runs a separate nnU-Net (pre-trained, then fine-tuned with a human in the loop) for dense organelles. This pipeline is all-SAM2 for now, with SAM2 fine-tuning planned rather than a second dense model.